

# Rakesh K M

Analog Layout Engineer — Custom IC Layout — AMS/RF Layout — Physical Verification (DRC/LVS) — Cadence Virtuoso

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## PROFESSIONAL SUMMARY

- M.Tech VLSI Design graduate with hands-on experience in full-custom Analog Layout using Cadence Virtuoso and Physical Verification using Calibre and Assura DRC/LVS.
- Completed industry-oriented Analog Layout Internship/Training using GlobalFoundries 40nm planar technology, with practical exposure to analog, mixed-signal and CMOS building blocks.
- Worked on full-custom layouts for a Folded-Cascode Low Noise Amplifier (M.Tech Thesis, SCL180nm) and Hybrid Full Adder (GPDK90nm), with hands-on exposure to TSMC 65nm Academic PDK, GlobalFoundries 40nm planar technology, and GPDK45/90/180 through custom analog layout projects and industry-oriented Analog Layout training.
- Skilled in device matching, common-centroid, interdigitation, dummy devices, guard rings, shielding, symmetry, fingering, manual routing, floorplanning, DRC/LVS, PEX and parasitic-aware layout optimization.

## TECHNICAL SKILLS

**Analog Layout:** Full-Custom Layout, Device Matching, Common-Centroid, Interdigitation, Dummy Devices, Guard Rings, Shielding, Symmetry, Fingering, Via Arrays, Manual Routing, Floorplanning, Layout Optimization

**Physical Verification:** Calibre DRC/LVS, Assura DRC/LVS, RC Parasitic Extraction (PEX), Post-Layout Simulation

**Process Technologies / PDKs:** GF40nm, TSMC65nm, SCL180nm, GPDK180nm, GPDK90nm, GPDK45nm

**EDA Tools:** Cadence Virtuoso, Calibre, Assura, Synopsys ICC2, Design Compiler, Synopsys VCS

**Analog / Mixed-Signal:** Differential Pair, Current Mirror, Single-Stage Op-Amp, Cascode Op-Amp, Comparator, Bandgap Reference, Level Shifter, DAC, Filters, Ring Oscillator, LDO Fundamentals

**Analog / RF Analysis:** S-parameters, Noise Figure (NF), IIP3, P1dB, K-factor,  $\mu$ -factor

**Programming:** C, Java, Python, TCL

## ANALOG LAYOUT INTERNSHIP / TRAINING

Semionics — Analog Layout Internship / Training

GlobalFoundries 40nm Planar

- Completed hands-on Analog Layout Internship/Training using Cadence Virtuoso, covering custom analog, mixed-signal and CMOS circuit implementation.
- Implemented differential pairs, current mirrors and passive structures using common-centroid, interdigitation, dummy devices, guard rings, shielding, symmetry and matching-oriented placement.
- Performed DRC/LVS, post-layout parasitic extraction (PEX), routing and layout optimization while following foundry design rules.
- Worked on CMOS blocks including standard cells, AOI gates, clocked NOR-based SR latch, CMOS D flip-flop, 8:3 priority encoder, Modulo-8 counter using JK flip-flops and 4-bit full adder; analog/mixed-signal blocks including single-stage and cascode Op-Amps, multi-voltage level shifter, 4/8-bit DACs, band-pass filter, CMOS bandgap reference, NMOS clamping circuits and ring oscillator.

## PROJECTS

Noise-Canceling Low Noise Amplifier (LNA)

2026

- Designed and analyzed CG-CS, cascode, and folded-cascode LNA topologies in SCL 180nm using Cadence Virtuoso
- Implemented full-custom analog layout using common-centroid placement, interdigitation, dummy devices, symmetry, guard rings and shielding for matching-critical devices.
- Optimized floorplanning, routing and device placement to minimize mismatch and parasitic effects while satisfying analog layout constraints.
- Performed Calibre DRC, LVS, post-layout RC parasitic extraction (PEX), and prepared layouts following foundry design rules and layout verification methodologies.

Low Power Hybrid Full Adder for Array Multiplier

2025

- Designed hybrid full-adder based 4-bit and 8-bit array multipliers in GPDK90nm CMOS using Cadence Virtuoso.
- Implemented transistor-level schematic and full-custom layout with parasitic-aware layout optimization for low-power operation.
- Performed Assura DRC, LVS and post-layout RC parasitic extraction (PEX).

## EDUCATION

Master of Technology (M.Tech) in VLSI Design

Vellore Institute of Technology, Chennai 2024–2026

Bachelor of Engineering (B.E.) in Electronics and Communication Engineering

Sri Sairam Engineering College, Chennai 2019–2023

## CERTIFICATIONS

NPTEL

- [System Design Through Verilog](#)
- [VLSI Design Flow: RTL to GDS](#)